Introduction to Wildlife Management and Conservation of Pakistan

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WILD LIFE MANAGEMENT AND CONSERVATION

COURSE OUTLINE:

- Wild life terminologies in management and conservation
- Wild life values
- Objectives of wildlife management
- Problems in wild life conservation and management
- Principals of reintroduction
- IUCN categories of wild life
- National Parks
- Wild Life Sanctuaries and Game reserves of Pakistan
- Management of National parks, sanctuaries, games reserves and related environmental problems
- Zoo rules
- Modern techniques for the control of environmental pollution in wild life areas
- Endangered species causes and measures for the conservation
- International conventions

INTRODUCTION

Wildlife:

Wildlife generally gives an expression large animals living in the forest such as, lions, elephants, wolves and animals that live in aquatic ecosystem such as fishes, crocodiles, whales etc.

Natural habitat of organisms is considered in wildlife.

Importance:

- Wildlife is an ecological part of our environment and its maintenance is essential for maintaining balanced ecosystem.
- Wildlife play an important role in the food chain converting the plant product into meat. Product into meat which is an important source of proteins.
- Wildlife indicates the health of environment around humans.
- It enhance esthetic of the environment.
- It offers unique opportunity for sport and healthy recreation.
- It serves as important medium for scientific research in environmental science, medicines, hunting, agriculture and other fields.
- Proper wildlife management can bring economic return through sport, hunting and wildlife tourism.
- Improved rural economy by generating employment opportunities through wildlife protection and development program wildlife tourism.
- Wildlife also provide job opportunities for the experts in the field.
- Wildlife tourism provide opportunities for improving relationship among people at National and International levels.
- Many wildlife animals act as scavengers (cleaner of the Earth surface/environment).
- Many wildlife species act as biological controls against crops in forest and plants agriculture.

Protected area in Pakistan:

Baluchistan = 15, *Azad Jammu Kashmir = 09,* *Khyber Pakhtunkhwa = 34,*

Punjab = 62, *Sindh = 50,* *Federal Area = 03,*

Northern Area = 19, **Total = 202**

National park:

Area owned by the Government set aside for the protection and preservation of its outstanding flora and fauna (plants and animals) in national state. It is accessible to the public for recreation, education and research activities.

National Parks in Pakistan:

Baluchistan =02, *Azad Jammu Kashmir = 01,* *Khyber Pakhtunkhwa = 03,*

Punjab = 02, *Sindh = 01,* *Federal Area = 01,*

Northern Area = 04, **Total = 14**

Wild Life Award:

Award is given to a person who has outstanding performance in the field of wildlife and award given in the form of cash and certificate appreciation.

Distribution among wildlife in Pakistan:

Mammals = 174, *Birds = 668,* *Reptiles = 177,* *Amphibians = 21,*

Fishes (Marine water) = more than 1,000, *Fishes (freshwater) 198,*

Insects = more than 5,000, *Flowering Plants = 5,700,* *Fungi = more than 4,500* *Algae = 775.*

Protected Area:

Area of land or sea especially dedicated for the protection and maintenance of biological diversity and of natural and associated cultural resources.

Causes of Wildlife:

Precaution of Human Beings:

- Hunting and shooting of wildlife animals for sports and trophies.
- Illegal and legal hunting for food.
- Illegal killing for skin, fur tasks, oil and decoration.
- Animal like many rabbits etc. are trapped live and soil to who need for medicinal experiment.

- In many countries some animals are trained such as monkey, bears and parrot for public shows.

Mode of Wildlife Conservation:

(i) Study of habitat:

To understand the habitats of wildlife animals.

(ii) Preservation:

To protect the natural habitat.

(iii)Wildlife Research:

- To find the status of each and every species.
- Food and feeding habitat and their place and food chain.
- Understand the biology and behavior of important species.
- To determine ecological environment of individual species.

(iv)Wildlife Legislation:

Hunting of endamaged species is strictly prohibited. No person is allowed to export and import of wildlife animal and its product.

(v) Wildlife Veterinary:

Veterinary should be made available to take care of animal against different types of diseases.

(vi)Involvement of International Agencies:

INCN, WWF help for the establishment and protect the endamaged species.

Act:

Hunting, capturing, killing are strictly prohibited in the area of wild life national park or in 3-miles away.

Firing of goons and other act which disturb the breeding place. Removal of plant or damaging or plant inside the area of national park is strictly prohibited.

NATIONAL PARKS

Protected area managed mainly for ecosystem protection and recreation which is defined as a natural area of land and/or sea, designated to

(a) Protect the ecological integrity of one or more ecosystems for present and future generations,
(b) Exclude exploitation or occupation inimical to the purposes of designation of the area and

(c)Provide a foundation for spiritual, scientific, educational, and recreational and visitor opportunities, all of which must be environmentally and culturally compatible

Definition

In 1969 the IUCN (International Union for the Conservation of Nature and Natural Resources) declared a national park to be a relatively large area with particular defining characteristics. A national park was deemed to be a place

- With one or several ecosystems not materially altered by human exploitation and occupation, where plant and animal species, geomorphological sites and habitats are of special scientific, educative and re-creative interest or which contain a natural landscape of great beauty.
- The highest competent authority of the country has taken steps to prevent or eliminate as soon as possible exploitation or occupation in the whole area and to enforce effectively the respect of ecological, geomorphological or aesthetic features which have led to its establishment.
- Visitors are allowed to enter, under special conditions, for inspirational, educative, cultural and re-creative purposes.

In 1971 these criteria were further expanded upon leading to more clear and defined benchmarks to evaluate a national park. These include:

- A minimum size of 1,000 hectares within zones in which protection of nature takes precedence
- statutory legal protection
- A budget and staff sufficient to provide sufficient effective protection
- Prohibition of exploitation of natural resources (including the development of dams) qualified by such activities as sport, fishing, the need for management, facilities, etc.

National parks of Pakistan

In Pakistan a National Park is an area of outstanding scenic merit and natural beauty where the landscape, flora and fauna are protected and preserved in a natural state. Public access for recreation, education and research is provided for. Access roads and other facilities should be planned so they do not conflict with the main objectives of national parks. Hunting wild animals is prohibited, as is firing gun or otherwise interfering with animals and plants. Clearing land for cultivation, mining or allowing polluted water to flow in National Parks is also prohibited. Under the regulations, these acts may be allowed for scientific purposes or to improve the park.

There are about 19 National Parks in Pakistan. Following are major National Parks of Pakistan:

- Margallah Hills National Park: Islamabad, established 1980, area 15,883 ha.
- Hazarganji Chiltan National Park: Balochistan, established 1980, area 15,555 ha.
- Hingol National Park: Balochistan, established 1997, area 6, 19,043 ha.
- Kirthar National Park: Sindh, established 1974, area 3, 08,733 ha.
- Ayubia National Park: Khyber Pakhtunkhwa, established 1984, area 3,122 ha.
- Chitral Gol National Park: Khyber Pakhtunkhwa, established 1984, area 7,750 ha.
- Sheikh Badin National Park: Khyber Pakhtunkhwa, established 1999, area 15,540 ha.
- Khunjerab National Park: Gilgit-Baltistan, established 1975, area 2, 27,143 ha.
- Deosai National Park: Gilgit-Baltistan, established 1993, area 3, 63,600 ha.
- Central Karakoram National Park: Gilgit-Baltistan, established 1993, area 9, 73,845 ha.
- Machhiara National Park: AJ & K, established 1996, area 13,532 ha.
- Lal Suhanra National Park: Punjab, established 1972, area 51,368 ha.
- Chinji National Park: Punjab, established 1987, area 6, 097 ha.
- Kala Chitta National Park: Punjab, established Dec. 2008, area 36,965 ha.
- Ghamot National Park:Azab jammu Kashmir,established2004,area 27394ha.
- Lake Lulu Sar National Park: NWFP, established2003, area 30375 ha.
- Lake Saiful Muluk: NWFP, eastablished2003, area 4867nha.
- Pir Lasora National Park:Azad jammu Kashmir,established2005,area5625ha.
- Toli Pir National Park: Azad Kashmir, established 2005, area5045ha.
- Shandur Hundrup National Park: Gilgit Baltistan established 1993,area 1,64,000 ha

Purpose and Objectives of National Parks

Protection of biodiversity is the main purpose for the establishment of a national park however, being biologically, geologically and culturally important, the national parks also serve to cater for education, recreation and scientific purposes for the public. People visit the parks to see and share the wonders of their land and to learn about the forces and the people who have shaped it through the centuries. Outstanding scenery, majestic places, natural flora and fauna in natural state renders positive impacts on visitors' mind thus helping divert a nation's attitude towards a healthy tract. National parks ensure the preservation of national natural heritage, culture and monuments, and present them to the public.

Role of National Parks in Nature Conservation

Due to the growing threats to nature like global warming, population expansion, habitat destruction and worldwide reduction of biological diversity, the national parks are now being considered as the ecological laboratories, gene pools and bulwarks against the irreversible change or the loss of species and hence, preserving the nature and ensuring the ecological health of the planet. National parks are the ideal places for eco-tourism, tracking, bird and animal watch and nature photography. National parks with their lush green forest components neutralize the carbon emissions in the surrounding environments.

WILD LIFE SANCTUARIES

It is a place or area where wild indigenous animals are kept in protection so they can breed and are thus preserved .Shooting or hunting is prohibited in this area. A sanctuary is established by a notification and it can be abolished by a similar process.

Wild life sanctuaries of Pakistan

Wildlife Sanctuary is an area set aside as an undisturbed preservation and breeding ground where public access is restricted or regulated beyond National Parks and Game Reserves. Whereas settlement and grazing by domestic livestock is allowed in a National Park, such activities are prohibited in Wildlife Sanctuaries.

The main objective of a Wildlife Sanctuary is to protect flora and fauna from extinction. Hunting is strictly prohibited in these areas. At present there are 99 Wildlife Sanctuaries in Pakistan, 66 officially recognized by and up to the standards of the International Union for Conservation of Nature.

Here are some of the bigger and more prominent ones.

Astore/Baltistan Wildlife Sanctuaries

Covering 800 sq km in total, these two areas are on the north side of the Astore Valley in Gilgit-Baltistan, east of Nanga Parbat, on either side of the Indus. They are prominent for being conservationist spots for the endangered Astore Markhor, a large species of wild goats unique to this region of the world, in addition to the endangered Snow Leopard, Brown Bear and the Lynx.

Cholistan Wildlife Sanctuary

The Cholistan desert area in the south eastern portion of the province of Punjab contains some of the most rare and interesting wildlife in Pakistan. Some of the rare animals of this region are the Desert Wolf, Indian fox, Red fox, Jackal, Indian Grey Mongoose, Nilgai Antelope, Saker Falcon, Black-backed Vulture and Saw-scaled Viper. An area of about 6,533 sq km has been declared as a Wildlife Sanctuary inside the desert.

Hub Dam Wild Life Sanctuary

The Hub Dam is Pakistan's third largest dam. It is situated in the north of Karachi bordering the Kirthar National Park. A large water storage reservoir was constructed in 1981 on the Hub River, in a region of arid plains and low stony hills. Much of the shoreline is steeply shelving and stony, but there are many shallow bays and small islands. The greater part of the reservoir (in Balochistan) is unprotected; the eastern shore and area south of the Dam (in Sindh) are protected in the Kirthar National Park and Hub Dam Wildlife Sanctuary, respectively. The lake formed by the dam covers an area of 70 sq km and is an ideal place for bird watching. There is plenty of waterfowl in the lake, both resident and migratory. The surrounding hills are home to the Urial, the Sindh Wild Goat, Chinkara Gazelle, the Pangolin and numerous other birds and reptiles.

Nara Desert Wildlife Sanctuary

The Nara desert provides refuge to a large number of wildlife in the Sindh province. The Nara Canal area, a chain of some 200 freshwater, brackish and saline lakes and marshes stretching for about 150 km along either side of the Nara Canal from Sorah in the north to Sanghar in the south, is known to be of great importance for wintering waterfowl and other wildlife.

The largest population of the endangered Mugger Crocodile in Pakistan is found here.

Chasma and Taunsa Barrage Dolphin Sanctuary

The Chashma and Taunsa Barrage are declared Wildlife Sanctuaries by the Punjab government. A very important breeding, staging and wintering area for a wide variety of waterfowl, regularly exceeding 20,000 birds, the area is also the biggest preservation spot of the endangered Indus Dolphin. Since the 1970s, the population of the Indus Dolphin has been stabilized and even significantly increased here.

Rann of Kutch Wildlife Sanctuary

The Rann of Kutch Wildlife Sanctuary lies in the southeast of the province Sindh near the border with India. The marshy Runn of Kutch and its surrounding desert area in Sindh is one of the best natural habitats for a numbers of animal and birds in the province. This area is known to be a breeding ground for flamingoes and a staging ground for Pelicans, Cranes, Storks and many species of waterfowl. This area also used to have the only population of the Indian Wild Ass in Pakistan. Unfortunately they are thought to have been exterminated.

GAME RESERVES

A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis. A private game reserve is an area of private land set aside by its owner for the same purpose as a game reserve. Parts of areas protected under some statutes may be exempted under pressure for agricultural extension or land development. Most of the areas have created to provide habitat protection for animal species commonly referred to as game (hunt able species for sport or meat).Providing protection for these species also offers a measure of security for many lesser known plant species and smaller animal species (IUCN, 1990)

Game reserves of Pakistan

Pakistan has declared 96 Game Reserves to be included in protected areas. But the conservation activities are not implemented properly Game reserves, in particular in private land, receive minimal protection due to the lack of legal provisions to control land use.

Wildlife protected areas and management issues

- Weakness in Law Enforcement
- Lack of Local Community's Involvement
- Deficiencies in Selection Criteria
- Limited Protected Area Categories
- Shortage of Professionals
- Lack of management Plans
- Lack of funds

Weakness in Law Enforcement

Existing wildlife laws do not provide an adequate framework for management. The laws give authority for protected area management to the provincial wildlife departments, but give no authority to these departments over the management of adjacent areas. Consequently, development activities in areas adjacent to protected areas often conflict with biodiversity conservation. There are a number of weaknesses related to enforcement of laws regarding conservation of biodiversity. With regard to the conservation of species, a serious weakness in the law is that it deals excessively with animal species with no provisions for the protection of threatened and endangered plant species. Attempts have been made to control the hunting of designated game animals, but most of these regulatory measures have proved difficult to enforce.

Lack of Local Community's Involvement

Current thinking on protected area management is that, to be effective, the communities living alongside the area should have a hand in management and should derive some benefits from the area. None of the existing categories of protected areas in Pakistan make allowance for participatory management by communities. In order to overcome these deficiencies a draft wildlife law empowering local communities to participate in joint wildlife management with

governments has been prepared and is currently under review by the provinces. Local communities rarely have any role in the management of protected areas. Little progress has been made on instituting collaborative management regimes for protected areas. Few efforts have been directed at raising public education and awareness in areas adjacent to protected areas, providing environmentally sound and sustainable development assistance to local communities, or formulating appropriate packages of incentives and disincentives. Consequently, local communities either continue to disregard protected area provisions leading to degradation of the protected area or, where those provisions are enforced against local communities' interests, conflicts have arisen. The conflict between local communities and park authorities in the Khunjerab National Park, arising from the loss f grazing rights, is well known. An attempt has been made to resolve and manage the conflict in Khunjerab, but a deep-rooted problem persists.

Deficiencies in Selection Criteria

The majority of Pakistan's protected areas were created in the 1970s, and paid insufficient attention to ecological criteria and the requirements of communities. Protected areas have been created haphazardly, often in the absence of any criteria for their selection, and boundaries drawn with little or no ecological basis. Besides, selection criteria for protected areas have also other serious shortcomings. No attention was being made for regional distribution of areas. Therefore considerable regional inequality in the distribution of Protected Areas across Pakistan can be observed. For example, whilst over 16% of Punjab is protected as one of three Protected Area categories (national park, wildlife sanctuary or game reserve); about 6% of the NWFP and less than 6% of Baluchistan is formally protected. (This is unfortunate; since these are the regions where most of Pakistan's remaining biodiversity is concentrated.

Limited Protected Area Categories

The three categories of Protected Areas used in Pakistan are too limited for contemporary needs. Most of the remaining unprotected areas of biodiversity significance are currently used and managed by local communities in one way or another. The best way to protect these areas will be through the establishment of collaborative management regimes. However, existing wildlife legislation in Pakistan does not provide for a protected area category in which sustainable use of resources and community involvement can take place (along the lines of IUCN Category VI: Managed Resource Protected Area). Moreover, many of the Protected Areas are too small and isolated to be effective. Most ecological zones are not adequately represented within the protected area system, including a majority of the critically threatened ecosystems. For example, there are no marine Protected Areas, very few coastal Protected Areas and no formal designation to protect the remaining juniper forests in Baluchistan.

Shortage of Professionals

There is lack of sufficient professionals in order to formulate and implement the policies regarding development and conservation of biodiversity. Untrained persons cannot successfully

manage the related problem and issues. Provincial wildlife departments lack the capacity to carry out their functions effectively, and in particular, suffer from a shortage of suitably trained personnel. Therefore management activities in remote areas of the country cannot be conducted in appropriate way. Moreover, current opportunities for training professionals in the area of conservation biology are very limited. There are institutions offering strong programs in forestry and agriculture, but there are no degree programs in wildlife management, bio-systematic, biodiversity conservation or community-based conservation of natural resources. A shortage of funding and lack of trained staff have also limited the amount of research on the identification, conservation, and sustainable use of biological diversity in Pakistan. There is little integration of research among institutions and disciplines, and very limited use of traditional knowledge in defining management programs.

Lack of management Plans

Most protected areas in Pakistan lack comprehensive management plans, and where plans do exist, they are not fully implemented. There is also a tendency to regard management plans as blueprints rather than adaptive strategies requiring constant updating. In recent years, some efforts have been made to redress this situation, and plans have been developed for the Margalla Hills National Park (Federal Territory) and Khunjerab National Park (Northern Areas). WWF Pakistan is also working on plans for the Kirthar National Park in Sindh and the Hazarganji Chiltan National Park in Balochistan. But there are still many National Parks without an adequate management plan. There is strong requirement to formulate management plans for these areas as soon as possible.

Lack of funds

There is lack of adequate funding in order to conduct management activities fruitfully. Although a reasonable amount is allocated by Government for these activities but it does not meet the needs required. The federal and provincial governments are the most important stakeholders with overall responsibility for providing an adequate policy and legal framework, enforcing regulations, building capacity and providing incentives and funds for the conservation of biodiversity. The policies and programs of key federal ministries i.e Environment, Local Government and Rural Development; Food and Agriculture; Finance; and Science and Technology and provincial departments of agriculture, livestock, forestry, wildlife and fisheries regarding funding are crucial to the conservation and sustainable use of biodiversity .

IUCN CATAGORIES OF WILD LIFE

The IUCN Red List Categories and Criteria are intended to be an easily and widely understood system for classifying species at high risk of global extinction. The general aim of the system is to provide an explicit, objective framework for the classification of the broadest range of species

according to their extinction risk. However, while the Red List may focus attention on those taxa at the highest risk, it is not the sole means of setting priorities for conservation measures for their protection.

Before 1994 the more subjective threatened species categories used in IUCN Red Data Books and Red Lists had been in place, with some modification, for almost 30 years. Although the need to revise the categories had long been recognized (Fitter and Fitter 1987), the current phase of development only began in 1989 following a request from the IUCN Species Survival Commission (SSC) Steering Committee to develop a more objective approach. The IUCN Council adopted the new Red List system in 1994.

OBJECTIVES OF IUCN

The IUCN Red List Categories and Criteria have several specific aims:

- to provide a system that can be applied consistently by different people;
- to improve objectivity by providing users with clear guidance on how to evaluate different factors which affect the risk of extinction;
- to provide a system which will facilitate comparisons across widely different taxa;
- to give people using threatened species lists a better understanding of how individual species were classified.

THE CATEGORIES

EXTINCT (EX)

A taxon is Extinct when there is no reasonable doubt that the last individual has died. A taxon is presumed extinct when exhaustive surveys in known and/or expected habitat, at appropriate times (diurnal, seasonal, annual), throughout its historic range have failed to record an individual. Surveys should be over a time frame appropriate to the taxon's life cycle and life form.

EXTINCT IN THE WILD (EW)

A taxon is Extinct in the Wild when it is known only to survive in cultivation, in captivity or as a naturalized
population (or populations) well outside the past range. A taxon is presumed Extinct in the Wild when exhaustive surveys in known and/or expected habitat, at appropriate times (diurnal, seasonal, annual), throughout its historic range have failed to record an individual. Surveys should be over a time frame appropriate to the taxon's life cycle and life form.

CRITICALLY ENDANGERED SPECIES (CR)

A category containing those species that possess an extremely high risk of extinction as a result of rapid population declines of 80 to more than 90 percent over the previous 10 years (or three generations), a current population size of fewer than 50 individuals, or other factors

ENDANGERED (EN)

A designation applied to species that possess a very high risk of extinction as a result of rapid population declines of 50 to more than 70 percent over the previous 10 years (or three generations), a current population size of fewer than 250 individuals, or other factors.

VULNERABLE (VU)

A category containing those species that possess a very high risk of extinction as a result of rapid population declines of 30 to more than 50 percent over the previous 10 years (or three generations), a current population size of fewer than 1,000 individuals, or other factors.

NEAR THREATENED (NT)

A taxon is Near Threatened when it has been evaluated against the criteria but does not qualify for Critically Endangered, Endangered or Vulnerable now, but is close to qualifying for or is likely to qualify for a threatened category in the near future.

LEAST CONCERN (LC)

A taxon is Least Concern when it has been evaluated against the criteria and does not qualify for Critically Endangered, Endangered, Vulnerable or Near Threatened. Widespread and abundant taxa are included in this category.

DATA DEFICIENT (DD)

A taxon is Data Deficient when there is inadequate information to make a direct, or indirect, assessment of its risk of extinction based on its distribution and/or population status. A taxon in this category may be well studied, and its biology well known, but appropriate data on abundance and/or distribution are lacking. Data Deficient is therefore not a category of threat. Listing of taxa in this category indicates that more information is required and acknowledges the possibility that future research will show that threatened classification is appropriate. It is important to make positive use of whatever data are available. In many cases great care should be exercised in choosing between DD and a threatened status. If the range of a taxon is suspected to be relatively circumscribed, and a considerable period of time has elapsed since the last record of the taxon, threatened status may well be justified.

NOT EVALUATED (NE)

A taxon is Not Evaluated when it is has not yet been evaluated against the criteria.

IUCN categories of threatened species

ENDANGERED SPECIES

An endangered species is a population of an organism (usually a taxonomic species), which because it is either few in number or threatened by changing environmental or predation parameters, leaving it at risk of becoming extinct. Many countries have laws offering special protection to these species or their habitats: for example, forbidding hunting, restricting land development or creating preserves. Only a few of the many endangered species actually make it to the official lists and obtain legal protection. Many more species become extinct, or potentially will become extinct, without gaining public notice. The greatest factor of concern is the rate at which species are becoming extinct within the last 150 years.

While species have evolved and become extinct on a regular basis for the last several hundred million years, the number of species becoming extinct since the Industrial Revolution has no precedent in biological history.
If this rate of extinction continues, or accelerates as now seems to be the case, the number of species becoming extinct in the next decade could number in the millions.
While most people readily relate to endangerment of large mammals or birdlife, some of the greatest ecological issues are the threats to stability of whole ecosystems if key species vanish at any level of the food chain.

The current rate of extinction is thought to be far greater than the expected natural rate, with many species going extinct before they have even been discovered. Shockingly, current estimates suggest that a third of the world's amphibians, a quarter of all mammals and one in eight birds are endangered.

Endangered species usually have a small or declining population size or a very limited range, meaning factors such as habitat loss, hunting, disease or climate change could cause them to disappear completely within our lifetimes.

The alarming rate at which species are disappearing is something which should be a cause for concern for us all. Not only do they add beauty and wonder to the natural world, they are also of great global economic importance. A great diversity of species maintains the ecosystems essential to our existence by helping to regulate our climate and by providing:

- clean air and water
- food
- medicines
- building and clothing materials
- fertile soils

With so many species at risk of extinction, the ARKive project is working to help raise the public profile of the world's endangered species, through the emotive power of wildlife films and photos. The ARKive project hopes to engage people with the natural world. We will only succeed in rescuing species from the brink of extinction if people know about their plight, importance and value.

CAUSES

Animals and plants face a large number of different **threats** with many of them being a direct result of human activity. Some of the most common threats include:

- **Habitat loss and habitat fragmentation** – The ever expanding human population constantly requires additional space and resources. Land is being cleared to harvest products such as timber as well as to make way for human settlement, agriculture and transport links.
- **Hunting and poaching** – A wide variety of animals have been hunted, or fished, beyond sustainable levels and now face possible extinction. Species, such as the tiger, are often hunted because they provide a resource such as food or parts which are used in traditional 'medicine'. However, some species, such as the cheetah, have been persecuted after gaining a negative reputation for feeding upon livestock or crops or posing a threat to human safety.
- **Invasive species** – Humans have introduced non-native species (both intentionally and accidentally) to a wide variety of habitats, often with devastating consequences. Introduced species may prove highly adaptable and competent than native species for resources. Introduced predators can decimate local species which are not adapted to avoid predation, for example ground dwelling birds like the kakapo.
- **Climate change** - Droughts, ocean acidification, the loss of sea ice and an increase in storms and extreme weather events can all threaten species' survival. Sedentary species like plants or specialist species which inhabit small ranges or islands, or those with specific habitat requirements are particularly vulnerable.
- **Disease** – Small populations, especially those which are limited in terms of genetic diversity are particularly vulnerable to disease. Disease can often be spread by domestic animals or accidentally introduced by humans travelling from an affected area to one which had not previously been exposed.
- **Collection / pet trade** – Many animals and plants, such as the Venus flytrap, have been collected from the wild beyond sustainable levels to be sold through the pet trade or be kept in private horticultural collections.
- **Pollution** – Acid rain, heavy metals, pesticides, plastic waste and oil spills all harm the environment and put species at risk. Chemicals are particularly harmful to species that live in water.

MEASURES

Conservation aims to protect the natural world and sustain biodiversity by carefully preserving and managing existing habitats and restoring areas which have been damaged or degraded.

Species conservation can also take place outside a species' natural habitat. For example, caring for an endangered animal in captivity, such as in a zoo, or preserving endangered plants through the use of seed banks.

In areas where humans and animals are competing for space or resources, particularly in poorer developing countries, it is important that conservation work takes into account the needs of local people and works alongside them in protecting their native species.

Some commonly used conservation actions are as follows:

- **Habitat preservation** – The ideal solution is to protect habitats before they are damaged. This can be achieved through the creation of national parks and marine protected areas. However, it is important to note that many larger species require extensive territories and designated protected areas may not be large enough to support them.
- **Habitat restoration** – Where a habitat has already been degraded it is sometimes possible to restore the habitat by carefully managing the land, removing invasive species and reintroducing native species that had been lost from the area. Some species are bred in captivity or relocated from other areas for this purpose.
- **Ex-situ conservation** – Many endangered species are bred in captivity to preserve their numbers and in some cases it is possible to reintroduce them to the wild. Some species, like the Golden arrow poison frog, have even been deliberately removed from the wild to protect them from the spread of disease and ensure that a small population is preserved. Plant species are often cultivated in nurseries and preserved via the use of seed banks.
- **Anti-poaching measures** – In remote areas guards are sometimes employed to protect endangered species, such as the mountain gorilla, from poachers. This can be a way of involving local communities in the protection of their wildlife whilst also providing some employment opportunities.
- **Wildlife corridors** – Where habitats have been fragmented by divisions such as roads, urban areas or farmland, populations become isolated and are unable to move throughout their natural range to find sufficient resources and mates. Wildlife corridors help to reconnect habitat fragments and maintain genetic diversity.
- **Laws and policies** – Some endangered species are protected by law or trade in them is restricted. CITES (The Convention on International Trade in Endangered Species) is an international agreement between governments to ensure that trade in wild animal and plant specimens does not threaten their survival.

INTERNATIONAL CONVENTIONS

Wildlife management and conservation transcends ecological systems because it take place simultaneously on land, in the water, and the air. Likewise, ecological systems transcend international boundaries, and therefore an action on one side can have a significant impact on the other, or even across several boundaries. Further, at international level, trade in wild plants and animals or their parts is known to have decimated populations of many species

Concerns at the international level over destruction of shared ecosystems, loss of biodiversity, and negative impacts on the environment in general have increasingly necessitated international

means of redress. Response has come in form of intergovernmental treaties or other agreements that constitute international environmental law. Such agreements govern cooperation among states on environmental matters of mutual interest or concern that one country cannot address alone. Often these agreements are between more than two countries and are hence referred to as multilateral environmental agreements (MEAs).

Some of the important international conventions are discussed below:

Antarctic Treaty System

The Antarctic Treaty System comprises a lot of different treaties that regulate the ways in which nations can interact with Antarctica. Since there is no native population of people that call Antarctica home, there is no technical "ownership" of the continent. In the past, this has caused a certain amount of concern within the international community as to the possibility of individual nations asserting ownership over the continent. The Antarctic Treaty System presents a framework upon which international relations can be based when it comes to issues regarding Antarctica. The major provisions in the treaty system are to promote free scientific investigation, and the banning of all military activity on the entire continent. Due to the strict levels of protection in place, interacting with wildlife or adversely affecting the environment of Antarctica is strictly forbidden.

Convention on Biological Diversity

This is a legally binding, international treaty whose focus is to promote biological diversity through a number of different methods. Primarily, the treaty seeks to reach its goals through conservation, sustainable use, and the equitable sharing of genetic biological resources. This basically means that the treaty is designed to help promote national strategies for conserving biological diversity, as well as for its sustainable use. The aspect of the treaty that most directly relates to ecotourism is the provision for the protection of local ecological knowledge and for the creation of educational initiatives. This has led to a growing amount of tourism opportunities throughout many of the areas that have signed and implemented this treaty. The United States of America is the only large developed country in the western world-- and one of only three recognized world nations-- that has not ratified the treaty.

Convention on International Trade in Endangered Species of Wild Fauna and Flora

The Convention on International Trade in Endangered Species of Wild Fauna and Flora, also known as "CITES", is a treaty which aims to protect wildlife from over-exploitation from international trade. The treaty provides different levels of protection for a large list of plant and animal species, working to protect their numbers in the wild. It does this by imposing a specialized permitting system on the transport and trade of specific listed species. By creating these extra hurdles, CITES has created an environment that has had a significant impact on the rampant over-exploitation of the species it monitors. The biggest effect that this has had on

ecotourism is that it has made the industry much more sustainable. Before the ratification of the treaty, trade in wild animals was largely regulated by local laws which can sometimes be abhorrently unproductive. By promoting the protection of exotic wild animals, CITES has led to a larger amount of biodiversity which has in turn made ecotourism more profitable. It is important for tourists to keep these regulations in mind whenever they find themselves in a position to buy animal parts or live animals in local bazaars.

Ramsar Convention

The Ramsar Convention is an international treaty that creates a structure through which nations can work together toward the conservation and sustainable use of wetlands. The treaty states a number of different social, recreational, and ecological functions that wetlands serve, and creates a number of provisions to help protect them from over-exploitation. The treaty is the largest international effort ever undertaken to promote the health, conservation, and study of wetland habitat. One of the largest impacts of this treaty has been on international water quality. Since wetlands act as filters for rainwater and runoff before they run to an aquifer, wetlands are an integral component of a sustainable water cycle. For tourists, a major benefit is not only the protection of genuinely wild landscapes, but also the protection of the many species of birds that rely on this healthy wetland habitat for their survival. For example, Canada alone has protected over 50,000 mi2 through its involvement in this program. This has led to a significantly lower impact on wild bird populations, and has in turn led to the continuation of the tourism industry that thrives around Canada's birding opportunities. The Brazilian Pantanal-- one of the largest wetlands in the world-- has also gained a significant level of protection from the Ramsar Convention.

The Bonn Convention

The objective of the convention on the Conservation of Migratory Species and Wild Animals (also known as the Bonn convention or CMS) is the conservation of wild animals throughout their range, with the emphasis on terrestrial, marine and avian migratory species. Migratory species are defined by the CMS as "the entire population or any geographically separate part of the population of any species or lower taxon of wild animals a significant proportion of whose members cyclically and predictably cross one or more national jurisdictional boundaries". The CMS or Bonn Convention was signed in 1979 in Bonn and entered into force on 1 November 1983. As of 1 January 2007 it includes 101 parties form Africa, Central and South America, Asia and Oceania.

IMPORTANT WET LANDS OF PAKISTAN

A wetland is a land area that is saturated with water, either permanently or seasonally. Wetlands are made up primarily of hydric soil, which supports aquatic plants. The water found in wetlands can be saltwater, freshwater, or brackish. There are four major kinds of wetlands, swamps, marshes, bogs and fens. Pakistan, despite having an arid climate, supports over 780,000 ha of wetlands covering 9.7% of the total land area, with 225 nationally significant wetlands . he high global significance of Pakistan's wetlands is attributable to the diversity of species that they support. In all, eighteen threatened species of wetlands dependent mammals are found here including the endemic Punjab Urial (Ovis vignei punjabiensis) and **Indus River Dolphin** (Plantanista minor), besides twenty threatened bird species and twelve reptiles and two endemic species of amphibians. Pakistan's wetlands also support between 191-198 indigenous freshwater fish species, including fifteen endemics and a total of 788 marine and estuarine fish species. The high altitude wetlands, characterized by sites such as Karumbar Lake, situated at an elevation of 4,150m, and Saucher Lake, at 4,250m on the **Deosai Plains**, represent a relatively unique category of alpine wetlands that is confined to the Himalaya, Hindukush and Karakoram mountain cordilleras.

List showing important Wetlands of Pakistan

No.	Name of Wetland	Status	Province	District/Area
1	Baghsar Lake	Not Protected	AJK	Bhimber
2	Banjosa	Not Protected	AJK	Rawalakot
3	Chatta Katha Lake	Not Protected	AJK	Muzaffarabad
4	Garhi Dupatta	Not Protected	AJK	Muzaffarabad
5	Mangla Lake	Not Protected	AJK	Mirpur
6	Nardi Lake	Not Protected	AJK	Muzaffarabad
7	Phoolawaei Lake	Not Protected	AJK	Muzaffarabad
8	Ratti Gali Lake	Not Protected	AJK	Muzaffarabad
9	River Neelum	Not Protected	AJK	Muzaffarabad
10	Saral Lake	Not Protected	AJK	Muzaffarabad
11	Akara Dam	Not Protected	Balochistan	Makran
12	Astola (Haft Talar) Island	Wildlife Sanctuary	Balochistan	Makran
13	Astola Sea Mount	Not Protected	Balochistan	Makran
14	Band Khushdil Khan	Game Reserve	Balochistan	Pishin
15	Boji Point	National Park (part of)	Balochistan	Lasbela
16	Chakhon	Not Protected	Balochistan	Zhob
17	Dasht Khor	Not Protected	Balochistan	Gwadar
18	Hamun-i-Lora	Not Protected	Balochistan	Chagai
19	Hamun-i-Mushkel	Not Protected	Balochistan	Chagai

20	Hanna Lake	Not Known	Balochistan	Quetta
21	Jiwani Estuary	Not Protected	Balochistan	Makran
22	Kalmat Khor	Not Protected	Balochistan	Makran
23	Marav Lake	Not Protected	Balochistan	Dera Bugti
24	Miani Hor	Not Protected	Balochistan	Lasbela
25	Pasni Bay	Not Protected	Balochistan	Gwadar
26	Qamardin Karez	Not Known	Balochistan	Qila Saifullah
27	Ras Malan	Not Protected	Balochistan	Lasbela
28	Siranda Lake	Not Protected	Balochistan	Lasbela
29	Spin Karez	Not Known	Balochistan	Quetta
30	Wasta Dam	Not Protected	Balochistan	Zhob
31	Zangi Nawar	Game Reserve	Balochistan	Chagai
32	Zhob River	Not Protected	Balochistan	Zhob
33	Rawal Lake	National Park (part of)	Federal Capital	Islamabad
34	Borith Lake	Not Protected	Northern Areas	Gilgit
35	Deosai Plateau	National Park	Northern Areas	Skardu
36	Gasho Lake, Sai Nullah	Not Known	Northern Areas	Gilgit
37	Hundrup Lake	Not Known	Northern Areas	Ghizer
38	Jutial Lake	Not Known	Northern Areas	Gilgit
39	Kachura Lake	Not Protected	Northern Areas	Skardu
40	Kharfaq Lake	Not Known	Northern Areas	Ghanche

These wetlands provide unparallel benefits and services, such as provisioning, food and fiber production, habitat for seasonal bird, regulating services such as water balance, groundwater recharge, flood mitigation and storm protection; cultural and social functions such as blessed and religious importance; providing recreation and tourism opportunities; and supporting functions such as soil formation and sediment maintenance. Moreover, it aesthetically doubles the value of that particular area. Hiking, boating, commercial fishing can be the source of remuneration for local resident.

Main threats to wetlands are the shortage of water, poor water quality, change in land use, encroachment and over-exploitation of natural resources, such as fish and wildlife. There is no public policy to maintain and manage this natural ecosystem. These are openly accessed for everyone and having no penalty on destruction. This is because of lack of public policy and

misconception. The problem is to define wetlands clearly. This will definitely help in understanding the importance of wetlands and the efforts being made to protect them.
- Despite all these, few efforts were made to protect this natural ecosystem by WWF, Ministry of Environment and treaty like Ramsar convention. The Ministry of Environment's Pakistan Wetlands Programme has been started. The Pakistan Wetlands Programme (PWP) aims to promote the sustainable conservation of freshwater and marine wetlands and their associated globally important biodiversity in Pakistan. The programme strategy is based on two sub-sets of objectives. The first will provide the required policy, institutional, technical and financial framework and generate positive public support essential for the mainstreaming of wetlands conservation. The second involves the design and implementation of Sustainable, participatory management plans for four independent Demonstration Sites, each chosen to be representative of a broad eco-region in Pakistan. It includes specific mechanisms to secure financial sustainability and enhanced replication and proliferation of viable wetlands management interventions in a nation-wide, on-going wetlands conservation initiative.

The Convention on Wetlands of International Importance, called the Ramsar Convention, is an intergovernmental treaty that provides the framework for national action and international cooperation for the conservation and wise use of wetlands and their resources. The Ramsar Convention is the only global environmental treaty that deals with a particular ecosystem. The treaty was adopted in the Iranian city of Ramsar in 1971 and the Convention's member countries cover all geographic regions of the planet.

INVENTORY OF WETLAND WATERFOWLS OF PAKISTAN

Bird fauna of Pakistan is predominantly Palearctic, especially in the winter time with an influx of migrant species. When it comes to the family Anatidae, Pakistan is home to many waterfowls, ducks, geese and swans. Few of them are native and majority is the migrant birds that stay in the moderate regions of Pakistan for breeding and to raise their young's. According to the zoologists, the distance covered by these birds during migration is around 4500 kilometers. They come from the colder regions of Russia, Siberia and Central Asia to the wetlands to Afghanistan, Karakorum range, across river Indus in Pakistan and finally towards India. During their journey, the birds make stopovers at lakes and water basins at Nowshera, Tanda Dam in Kohat, Swat, Chitral, and Punjab and at Haleji, Keenjaar and Lungsee lakes in Sindh. Likewise, Nara Canal in Khairpur and Drigh Lake in Larkana are also some noteworthy and excellent bird watching sites in the province of Sindh. Water birds inhabit or temporarily use wetlands because of the diversity of microhabitats for feeding, nesting and resting, as well as food richness in the water. Such habitats are among the highly complex, diverse and biologically productive ecosystem in the biosphere Birds act as bio indicator for the environment of the area.

Table 1: Inventory of wetland water fowls of Mangla Dam

Common names	Scientific names
Chukar	*Alectoris chukar*
Common Quail	*Coturnix coturnix*
Rain Quail	*Coturnix coromandelica*
Small Buttonquail	*Turnix sylvatica*
Grey Francolin	*Francolinus pondicerianus*
Black Francolin	*Francolinus francolinus*
Greylag Goose	*Anser anser*
Bar headed goose	*Anser indicus*
Greater White Fronted Goose	*Anser albifrons*
Common Shelduck	*Tadorana tadorana*
Ruddy_Shellduck	*Tadorna Ferruginea*
Gadwall	*Anas strepera*
Eurasian Wigeon	*Anus Penelope*
Mallard	*Anas platyrhynchos*
Spot_Billed Duck	*Anas poecilorhyncha*
Common Teal	*Anas crecca*
Garganey	*Anas querquedula*
Northern Pintail	*Anas Acuta*
Northern Shoveler	*Anas clypeata*
red_crested pochard	*nettta rufina*
Common Pochard	*Aythya ferina*
Ferruginous Pochard	*Aythya nyroca*
Tufted Duck	*Aythya fuligula*
Common Merganser	*Mergus merganser*
Eurasian Wryneck	*Jynx torquilla*
Yellow-Crowned Woodpeaker	*Dendrocopos mahrattensis*
Black_rumped Flameback	*Dinopium benghalense*
Coppersmith Barbet	*Megalaima haemacephala*
Indian Gray Hornbill	*Ocyceros birostris*
Common Hoopoe	*Upupa epops*
European Roller	*Coracias garruls*
Indian Roller	*Coracias benghalensis*
Common Kingfisher	*Alcedo atthis*
White-Throated Kingfisher	*Halcyon smyrnensis*
Crested Kingfisher	*Megaceryle lugubris*
Pied kingfisher	*Ceryle rudis*
Green Bee-eater	*Merops orientalis*

Blue-cheeked Bee-eater	*Merops persicus*
Blue-Tailed Bee-eater	*Merops philippinus*
Pied cuckoo	*Clamator Jacobinus*
Common hawk Cuckoo	*Hierococcyx varius*
Indian Cuckoo	*Cuculus Micropterus*
Asian Koel	*Eudynamys Scolopacea*
Greater Coucal	*Centropus Sinensis*
Alexadrine Parakeet	*Psittacula eupatria*
Rose -ringed Parakeet	*Psittacula krameri*
Plum-Headed Parakeet	*Psittacula cyanocephala*
House Swift	*Apus affinis*
Mountain scops owl	*Otus spilocephalus*
Pallid scops owl	*Otus brucei*
Collared scops owl	*Otus Bakkamoena*
Collared Owlet	*Glaucidium brodiei*
Asian barred owlet	*Glaucidium cuculoides*
Spotted Owlet	*Athene brama*
Dusky eagle owl	*Bubo coromandus*
Short-eared owl	*Asio flammeus*
Eurasian Nightjar	*Caprimulgus europaeus*
Savanna NightJar	*Caprimulgus affinis*
Rock Pigeon	*Columba livia*
Oriental turtle dove	*streptopelia orientalis*
Laughing dove	*Streptopelia senegalensis*
Spotted Dove	*Streptopelia chinensis*
Red Collared dove	*Streptopelia tranquebarica*
Eurasian collared dove	*Streptopelia decaocto*
Yellow_footed green pigeon	*Treron phoenicoptera*
Demoiselle Crane	*grus virgo*
Common Crane	*grus grus*
Purple Swamphen	*Porphyrio Porphyrio*
Water rail	*Rallus Aquaticus*
Baillan's Crake	*Porzana pusilla*
Spotted crake	*porzana porzana*
Ruddy_Breasted crake	*Porzana fusca*
Common Moorhen	*Gallinula chloropus*
Common Coot	*Fulica atra*
chestnut_bellied sandgrouse	*Pterocles exustus*
Common Snipe	*Gallinago gallinago*
Jack Snipe	*Lymnocryptes minimus*
Greater painted_Snipe	*Rastratula benghalensis*

Eurasian Curlew	*Numenius arquata*
Spotted Redshank	*Tringa erythropus*
Common Redshank	*Tringa totanus*
Marsh Sandpiper	*Tringa Stagnatilis*
Common Green Shank	*Tringa Nebularia*
Green Sandpiper	*Tringa ochrobus*
Wood sandpiper	*Tringa glareola*
Common Sandpiper	*Actitis hypoleucos*
Little Stint	*Calidris minuta*
Temmink's stint	*Calidris temminckii*
ruff	*Philomachus pugnax*
Pheasant_ tailed Jacana	*Hydrophasianus chirurgus*
Black-winged stilt	*Himantopus himantopus*
Pied Avocet	*Recurvirostra avosetta*
Grey Plover	*Pluvialis Squatarola*
Little Ringed Plover	*Charadrius dubius*
Kentish Plover	*Charadrius alexandrinus*
Northern Lapwing	*Vanellus vanellus*
Red_Wattled Lapwing	*Vanellus indicus*
White-Tailed lapwing	*vanellus leucurus*
Indian Courser	*Cursorius Coromandelicus*
small pratincole	*glareola lactea*
caspian gull	*larus cachinnans*
Brown-Headed Gull	*Larus brunnicephallus*
Black headed gull	*larus ridibundus*
Gull_billed tern	*Gelochelidon nilotica*
River Tern	*Sterna aurantia*
Common Tern	*Sterna hirundo*
Black-Bellied tern	*Sterna acuticauda*
whiskered tern	*Chlidonias leucopterus*
osprey	*pandion haliaetus*
Black_shouldered kite	*Elanus Caerusleus*
Black Kite	*Milvus migrans*
Brahminy Kite	*Haliastur indus*
Pallas's Fish Eagle	*Haliaeetus leucoryphus*
Lammergeier	*Gypaectus Barbatus*
Egyptian vulture	*Neophron percnopterus*
White_Rumped vulture	*Gyps bengalensis*
Eurusian griffon	*Gyps fulvus*
Short-Toed Snake eagle	*Circaetus gallicus*
Eurasian Marsh Harrier	*Circus Aeruginosus*

Hen Harrier	*Circus cyaneus*
Pallid harrier	*circus macrourus*
Shikra	*Accipiter badius*
Eurasian Sparrow Hawk	*Accipiter nisus*
Northern Goshawk	*accipiter Gentilis*
White-Eyed Buzzard	*Butastur teesa*
Common Buzzard	*Buteo Buteo*
Long-Legged Buzzard	*Buteo rufinus*
Tawny Eagle	*Aquila rapax*
Steppe Eagle	*Aqulia nipalensis*
Bonelli's Eagle	*Hieraaetus fasciatus*
Common kestrel	*Falco tinnunculus*
red necked falcon	*falco chicquera*
merlin	*falco columbarius*
Little grebe	*Tachybaptus Ruficollis*
Great Crested Grebe	*Podiceps Cristatus*
Black_neckd grebe	*Podiceps nigricollis*
Darter	*Anhinga melanogaster*
Little Cormorant	*Phalacrocorax niger*
Indian Carmorant	*Phalacrocorax Fuscicollis*
Great Cormorant	*Phalacrocorax Carbo*
Litle Egret	*Egretta garzeeta*
Grey Heron	*Ardea cinerea*
Purple Heron	*Ardea purpurea*
Great Egret	*Casmerodius Albus*
Intermediate Egret	*Mesophoyx intermedia*
Cattle Egret	*Bubulcus ibis*
Indian pond Heron	*Ardeola grayii*
Black_crowned night heron	*Nycticorax nycticorax*
Cinnamon Bittern	*Ixobrychus cinnamomeus*
Great Bittern	*Botaurus Stellaris*
Eurasian Spoonbill	*Platalea leucorodia*
Black Stork	*Ciconia nigra*
red backed shrike	*Lanius collurio*
Rufous-Tailed Shrike	*Lanius isabellinus*
long tailed shrike	*lanius schach*
southren grey shrike	*lanius meridionalis*
Rufous Treepie	*Dendrocitta Vagabunda*
Red_Billed chough	*Pyrrhocorax pyrrhocorax*
Yellow-Billed Chough	*Pyrrhocorax graculus*

House Crow	*Corvus splendens*
Common Raven	*Corvus Corax*
Eurasian Golden oriole	*Oriolus Oriolus*
Small Minivet	*Pericrocotus cinnamomeus*
Long-Tailed Minivet	*Pericrocotus ethologus*
white-Browed Fantail flycatcher	*Rhipidura aureola*
Black Drongo	*Dicrurus Macrocercus*
Asian Paradise_Flycatcher	*Terpsiphone paradisi*
common woodshrike	*tephrodornis pondicerianus*
Brown Dipper	*Cinclus pallasii*
Blue Rock Thrush	*Monticola solitarius*
Blue whistling thrush	*Myophonus caeruleus*
Plain_Backed Thrush	*Zoothera Mollissima*
Grey_winged blackbird	*Turdus boulboul*
Chestnut Thrush	*Turdus Rubrocanus*
Dark-Throated Thrush	*Turdus ruficollis*
Dark-sided flycatcher	*Muscicapa sibirica*
Red-throated Flycatcher	*Ficedula parva*
Kashmir flaycatcher	*Ficedula subrubra*
verditer flycatcher	*eumyias thalassina*
blue throated flycatcher	*cyornis ruberculoides*
Bluethroat	*luscinia svecica*
Indian Blue Robin	*Luscinia Brunnea*
Oriental Magpie Robin	*Copsychus saularis*
Indian robin	*Saxicoloides fulicata*
Rufous Backed Redstart	*Phoenicurus erythronota*
blue_capped redstart	*phoenicurus coeruleocephalus*
Black Redstart	*phoenicurus ochruros*
plumbeous water redstart	*Rhyacornis fuliginosus*
White-Bellied Redstart	*Hodgsonius Phaenicuroides*
Common Stonechat	*Saxicola Torquata*
Pied Bushchat	*sexicola Caprata*
Varriable Wheatear	*Oenanthe picata*
Brahminy starling	*sturnus pagodarum*
common starling	*sturnus vulgaris*
common Myna	*Acridotheres tristis*
Bank Myna	*Acridotheres ginginiaianus*
Jungle Myna	*Acridotheres fuscus*
Kashmir Nuthatch	*Sitta cashmirensis*
white-cheeked Nuthatch	*Sitta Leucopsis*

Wallcreeper	*Tichodroma Muraria*
Bar-Tailed Treecreeper	*Certhia himalayana*
Great Tit	*Parus Major*
Black-throated Tit	*Aegithalos Concinnus*
White-Throated Tit	*Aegithalos niveogularis*
Pale Martin	*Riparia diluta*
Plain Martin	*Riparia paludicola*
Barn Swallow	*Hirundo rustica*
wire-tailed Swallow	*Hirundo smithii*
streak_throated swalllow	*hirundo fluvicola*
Asian house Martin	*Delichon dasypus*
Red-vented Bulbul	*Pycnonotus Cafer*
Rufous-vented Prinia	*Prinia Burnesii*
Rufous Fronted Prinia	*Prinia buchanai*
Yellow Bellied Prinia	*Prinia Flaviventris*
Ashy Prinia	*Prinia Socialis*
Plain prinia	*Prinia inornata*
Graceful Prinia	*Prinia gracilis*
Oriental white-eye	*Zosterops palpebrosus*
Common Tailorbird	*Orthotomus sutorius*
Greater whitethroat	*Sylvia Communis*
Lesser whitethroat	*sylvia curruca*
Grey Sided Bush Warbler	*Cettia Brunnifrons*
Moustached Warbler	*Acrocephalus Melanopogon*
Paddyfiled Warbler	*Acrocephalus agricola*
Blyth's Reed werbler	*Acrocephalus drumetorum*
Clamorous Reed Warbler	*Acrocephalus stentoreus*
Booted Warbler	*Hippolais caligata*
Common Chiffchaff	*phylloscopus collybita*
Plain Leaf warbler	*Phylloscopus neglectus*
Tickell's leaf warbler	*Phylloscopus affinis*
Hume's Warbler	*Phylloscopus humei*
Greensish warbler	*Phylloscopus Trochiloides*
Tytler's Leaf Warbler	*Phylloscopus tytleri*
Western crowned warbler	*phylloscopus occipitalis*
yellow-eyed Babbler	*Chrysomma sinense*
Common babblar	*Turdoides Caudatus*
Striated Babbler	*Turdoide earlei*
Jungle babbler	*Turdoides Striatus*
singing bushlark	*mirafra cantillans*
Ashy-crowned sparrow lark	*Eremopterix grisea*

Bimaculatcd lark	*Melanocorypha bimaculata*
sand lark	*Calandrella raytal*
Crested Lark	*Galerida cristata*
Eurassian Sky Lark	*Alauda Arvensis*
Oriental sky lark	*Alauda gulgula*
Purple sunbird	*Nectarinia asiatica*
House Sparrow	*Passer domesticus*
Whilte Wagtail	*Motacilla alba*
white browed wagtail	*Motacilla Maderaspatensis*
Citrine wagtail	*Motacilla citreola*
yellow wagtail	*Motacilla Flava*
Grey wagtail	*Motacilla cinerea*
Paddyfield pipit	*Anthus rufulus*
Tawny Pipit	*Anthus campestris*
upland pipit	*Anthus sylvanus*
Tree pipit	*Anthus Trivivialis*
Rossy Pipit	*Anthus Roseatus*
Water Pipit	*Anthus Spinoletta*
Streaked Weaver	*Ploceus Manyar*
Baya Weaver	*Ploceus Philippinus*
Common Rosefinch	*Carpodacus erythrinus*
Rock bunting	*Emberiza cia*
Red Headed Bunting	*Emberiza Bruniceps*
Reed Bunting	*Emberiza Schoeniclus*
great grey shrike	*lanius excubitor*
crested serpent eagle	*spilornis cheela*
red necked grebe	*podiceps grisegena*

Threats to the Waterfowls of Pakistan

The environment for these waterfowls in Pakistan is becoming endangered for which the worldwide wildlife experts have growing concerns. Wildlife experts believe that some 80 per cent decline has been witnessed in the population of migratory birds in Punjab province in recent years mainly due to vanishing of forests, hunting, poaching, environmental pollution and massive agricultural activities. The UN has also warned that 84 per cent of migratory birds have the potential to be affected by climate change. Lowered water tables, changes in food supplies and prey range, rising sea levels, and increased storm frequency are the biggest threats to birds. The authorities have to take serious notice as the situation is quite alarming for these waterfowls and other migratory birds in Pakistan. They contribute a lot to the ecology of the country and their resting areas are the major tourist attraction as well.

ENVIROMENTAL POLLUTION IN WILDLIFE AREA AND MODERN TECHNIQUES FOR CONTROL

Pollution is one of the primary ways in which humans have caused drastic modifications of wildlife habitat. Historically we have regarded the air, water, and soil that surround us as waste receptacles and have given little consideration to the ecological consequences of our actions. As a result, wildlife populations are confronted with a bewildering array of pollutants that we release into the environment either by intent or accident.

In some cases wildlife populations have suffered severe losses or even faced extinction due to pollution. For example, the bald eagle, peregrine falcon, and brown pelican all nearly became extinct before scientists discovered that the synthetic chemical DDT was the cause of devastating reproductive failure in these species. Oil spills, such as the fouling of the coast of southern Alaska by the grounding of the Exxon Valdez, take an immediate toll on many species with the misfortune of living near such blunders. Toxic metals can kill adult members of wildlife populations and cause the production of deformed offspring, as seen at Kesterson Reservoir in the San Joaquin Valley. Acid rain has caused hundreds of fish populations to disappear from lakes in the northeastern U.S. and Scandinavia. In this chapter each of these notorious instances of the impacts of pollution on wildlife are described. The chapter also provides a general discussion of the origins and effects of synthetic chemicals, oil spills, toxic metals, and acid rain. Chronic and frequent noise interferes with animals' abilities to detect important sounds, whereas intermittent and unpredictable noise is often perceived as a threat. Importantly, these effects can lead to fitness costs, either directly or indirectly.

Techniques for controlling air pollution

Integrated pest management

The excessive use of Insecticides and pesticides in solid, liquid and aerosols sprays have posed serious impacts on the life sustaining on earth. Integrated Pest Management (IPM) ideally combines biological and cultural controls with limited pesticide use to keep pest populations below economically damaging levels, prevent future pest problems, and minimize the harmful effects of pesticides on wildlife.

Beneficial insects, birds, and mammals are natural enemies of many crop pests and can play an important role in IPM. Landowners and managers spend a considerable amount of time and money to control pest populations where natural pest inhibitors are lacking. Modified farming techniques, increased crop diversity, and use of cover crops and conservation buffers can increase food and cover on croplands for many species, including those beneficial in controlling pest populations. Birds are another recently recognized addition to the list of biological pest control agents. When used in combination with other pest control treatments, birds may help

reduce populations of insects and small mammals. Erecting perches and artificial nesting structures for raptors and songbirds is an easy way to complement IPM efforts. Perches and nesting structures can be placed around the perimeter of crop fields or in nearby suitable habitat.

Oil spilling and control measures

The consequences of oil exposure to marine and coastal wildlife include both immediate impacts on health as well as long-term changes in behavior. In small doses, the presence of oil can cause temporary physical harm to animals. Types of trauma can include skin irritation, altering of the immune system, reproductive or developmental damage, and liver disease. When large quantities of oil enter a body of water, chronic effects such as cancer, and direct mortality of wildlife becomes more likely.

Direct effects of oil spills on wildlife

Oil spills can impact wildlife directly through three primary pathways:

Ingestion - when animals swallow oil particles directly or consume prey items that have been exposed to oil

Absorption - when animals come into direct contact with oil

Inhalation - when animals breathe volatile organics released from oil or from "dispersants" applied by response teams in an effort to increase the rate of degradation of the oil in seawater.

Ingestion of oil or dispersants can cause gastrointestinal irritation, ulcers, bleeding, diarrhea, and digestive complications. These complications may impair the ability of animals to digest and absorb foods, which ultimately leads to reduced health and fitness. Ingestion can occur at multiple levels of the food chain. Herbivorous (plant-eating) wildlife, such as sea turtles, may consume vegetation that has been coated with oil particles. Carnivorous (animaleating) wildlife, such as shorebirds that feed on clams, mussels, or worms buried in the intertidal area, may consume prey organisms that have been exposed to oil sediments washed onto the shoreline. Baleen whales (those with hair-like teeth used to trap small particles from the ocean water) can become incapacitated when oil clogs their filtering device – in extreme cases this fouling of the baleen can lead to starvation and death. Top predators may become vulnerable to large quantities of pollutants through bioaccumulation (the increased concentration of toxins found at higher level of the food chain).

Absorption of oil or dispersants through the skin can damage the liver and kidneys, cause anemia, suppress the immune system, induce reproductive failure, and in extreme cases cause death. The skin of some species may experience irritation, burns, and infections. Exposure of oil to fish eggs in the water or sea turtle eggs on beaches can lead to slower embryonic growth, lower hatching rates, and developmental impairments.

Inhalation of volatile chemicals (vaporized materials released by oil floating on the surface) commonly occurs among those species of wildlife that need to breath at the surface of the water. Inhalation of these harmful materials can cause respiratory inflammation, irritation, emphysema, or pneumonia. Manatees, dolphins, whales, and sea turtles all come to the surface to breathe periodically, and are susceptible to this risk.

Indirect effects of oil spills on wildlife

Oil spills can also have indirect effects on wildlife by causing changes in behavior: relocation of home ranges as animals search for new sources of food increases in the amount of time animals must spend foraging disruptions to natural life cycles.

Eutrophication

Algae differ from microscopic animal life in our water bodies in their mode of respiration: they release more oxygen during the day than they use, and absorb more carbon dioxide than they release, while animal and other non-photosynthetic organisms release carbon dioxide and absorb oxygen from their environment. Algae usually react in an opposite manner during the night, when they act as dead organic matter adding BOD load. It is important to carefully consider any action of removal of algae from a body of water: oxygen supplied by algae during photosynthesis in beneficial to most forms of life in streams, thus the removal would often be a detriment rather than a benefit.

The main effects caused by eutrophication can be summarized as follows :

1. Species diversity decreases and the dominant biota changes

2. Plant and animal biomass increase

3. Turbidity increases

4. Rate of sedimentation increases, shortening the lifespan of the lake

5. Anoxic conditions may develop

Because of the high concentration of organisms in a eutrophic system, there is often a lot of competition for resources and predator pressure. This high degree of competition and the sometimes-high chemical or physical stress make high the struggle for survival in eutrophic systems. As a result the diversity of organisms is lower in eutrophic than in oligotrophic systems.

There are several measures that can be put in place to maintain the health of our water bodies, without severely impacting our way of life:

Riparian Buffers: These are vegetated areas between a river and adjacent areas of land use, and are a widely used method for removing pollutants (like the excess nutrients in eutrophication)

from agricultural areas. This means less surface run-off meets a stream/river, leading to less nutrients being deposited at their mouth (e.g. in a lake or sea), and some prevention of eutrophication. Lee, Isenhart and Schultz (2003) found a combined species buffer zone can remove between 80-97% of the nitrogen and phosphorous run-off. However different species have been found to contribute different rates of absorption to a buffer zone (Haycock and Pinay, 1993). Older forests have also been shown to be much less efficient at trapping N and P than younger forests (Mander et al., 1997).

Maintenance of river floodplains: Maintaining flat plains adjacent to rivers can also help to reduce nutrient-load in mouth water bodies; as when the river breaks its banks and floods, sediments and nutrients are deposited on the plains either side of the river. Tockner et al., 2002, found of a river-floodplain system in Switzerland that the floodplain served as a major sink of phosphorus and suspended matter, and was never a source for organic nutrients. However quantification of their use in preventing eutrophication does vary; Tockner et al., 1999, found of a system in Austria that the floodplain did serve as a sink for sediments and nitrates, but was actually a source for dissolved organic carbon and algal biomass.

Divertion of waste pipes: A fairly obvious measure, sewage pipes that offload into water bodies should be diverted away from areas in danger of eutrophication. Alternatively, wastewater can be treated to reduce the amount of nutrients carried into water bodies. This can dramatically decrease incidences of eutrophication, as anthropogenic waste is one of the widest known causes for destruction of lake ecosystems (Sreenivasan, 1969).

These are all ways in which we can fairly easily reduce anthropogenic pollution and associated eutrophication of water bodies. While methods such as re-meandering and creation of floodplains may have high monetary costs and require maintenance, these are still preferable to the difficult, time-consuming, and often impossible process of recovering systems damaged by unnatural eutrophication.

Deforestation and afforestation

Deforestation is a particular concern in tropical rainforests because these forests are home to much of the world's biodiversity. For example, in the Amazon around 17% of the forest has been lost in the last 50 years, mostly due to forest conversion for cattle ranching. Deforestation in this region is particularly rampant near more populated areas, roads and rivers, but even remote areas have been encroacheUndernourishment leading to death. Since herbivorous animals depend on plants for their survival, hence cutting down of forest could make them malnourished that lead to their death. In addition, there are carnivorous animals who are dependent on other animals are undergoing the problem of malnourishment.

Shifting of scores of wildlife animals to urban areas: Huge scale deforestations m wild animals to spread to the urban and rural areas, which lead to their death due to human-carnivore encounters. The number of attacks by lions and leopards on the human has been increased in past ten years. Many times, these encounters are accidental when wild animals take shelter in the sugarcane field. Besides, animosity between human and wild animals is also due to the loss of livestock.

Greenhouse effect: Carbon dioxide is a greenhouse gas, which let heat from the sun to protrude and hinders its escape from the atmosphere, which is required for maintaining a warm temperature on earth. However, the burning of fossil fuels and other activities are increasing greenhouse gases which are further raising the temperature of the earth, and this is called as the greenhouse effect.

Forests are biosphere's vital part for controlling atmospheric carbon dioxide by utilizing it for photosynthesis. However, people are clearing the forests for monetary reasons, which have led to an increase of carbon dioxide levels in the atmosphere.

Many nature and wildlife lovers have started taking efforts to save the various wild organisms on the verge of annihilation and the world from running out if its verdant vegetation. One has to understand the importance of wildlife, enriching the natural exquisiteness of the geography. It would really be an ignominy for anybody destroying their home.

The best solution to deforestation is **to curb the felling of trees, by employing a series of rules and laws to govern it**. Deforestation in the current scenario may have reduced however it would be too early to assume. The money-churner that forest resources can be, is tempting enough for deforestation to continue.

2. Clear cutting of forests must be banned. This will curb total depletion of the forest cover. It is a practical solution and is very feasible.

3. Land skinned of its tree cover for urban settlements should be urged to plant trees in the vicinity and replace the cut trees. Also the **cutting must be replaced by planting young trees to replace the older ones that were cut.** Trees are being planted under several initiatives every year, but they still don't match the numbers of the ones we've already lost.

Maintenance of carrying capacity

Carrying Capacity: the population that can be supported indefinitely by its supporting systems.

In ecological terms, the carrying capacity of an ecosystem is the size of the population that can be supported indefinitely upon the available resources and services of that ecosystem. Living within the limits of an ecosystem depends on three factors:

- The amount of resources available in the ecosystem
- The size of the population
- The amount of resources each individual is consuming

he planet's carrying capacity is the number of people that can live on it without dangerously threatening its future. Given rapid population growth on a planet that is ultimately finite, and our present inability to meet even the basic needs of much of the world's population, the human carrying capacity of the Earth has become a significant issue. The planet could support many more people living a simple rural village life than an American suburban lifestyle. One measure that has been developed to estimate carrying capacity and the share of each person is the ecological footprint.

The idea of carrying capacity relates closely to that of sustainable development, because both refer to the need to live off of interest rather than capital. If we cut down forests faster than they can grow back, intensify agriculture until it robs the soil of its fertility, and make profligate use of non-renewable resources such as minerals and fossil fuels, we can increase our standard of living and/or the number of people living on the planet, but we reduce its capacity to support people in the future. The success of development in the West has come partly from diminishing stocks of non-renewable resources, and partly from the import of cheap primary commodities often produced by over-exploitation in developing countries, just as the empires of earlier times flourished through colonial exploitation.

Noise pollution

As human beings continue their encroachment upon the last remaining vestiges of untouched wilderness, wildlife populations around the globe continue to diminish in size. The impacts of human encroachment and environmental pollution are evident wherever research biologists perform their studies: loss of habitat and territory; loss of food supply; behavioral changes in mating predation and migration; and changes in interspecies relationships, altered predator-prey balance, increased competition for food and shelter.

Human-induced noise pollution is one of many factors contributing to the depletion of wildlife populations. Laboratory studies and limited field research have uncovered four major ways in which animals are adversely affected by noise pollution:

- Hearing loss, resulting from noise levels of 85 db or greater;
- Masking, which is the inability to hear important environmental cues and animal signals;

- Non-auditory physiological effects, such as increased heart rate and respiration and general stress reaction; and
- Behavioral effects, which vary greatly between species and noise characteristics, resulting in, for example, abandonment of territory and lost reproduction.

Plant a tree, or two - According to the USDA National Agroforestry Center, "Planting "noise buffers" composed of trees and shrubs can reduce noise five to ten decibels (reduces noise approximately 50% to the human ear). To achieve this effect, the species and the planting design must be chosen carefully. Generalized recommendations to reduce noise with rows of trees and shrubs include, planting the noise buffer close to the noise source (rather than close to the area to be protected), planting trees/shrubs as close together as the species will allow and not be overly inhibited and using plants with dense foliage. A diversity tree species, with a range of foliage shapes and sizes within the noise buffer may also improve noise reduction. Foliage of the plants should persist from the ground up. A combination of shrubs and trees may be necessary to achieve this effect. Evergreen varieties that retain their leaves will give better year-round protection. When possible use taller plants. Where the use of tall trees is restricted, use combinations of shorter shrubs and tall grass or similar soft ground cover as opposed to harder paved surfaces.

Light pollution

Animals and plants live by a rhythm which is attuned to our planet's 24-hour cycle. !is is an inherited trait, which is passed on through the genes of a species. Humans may

notice a change in their circadian rhythm when they travel by airplane between several time zones, characterized by sleepiness, lethargy, or a general sense that something is "off." Wildlife and fish experience this same disorientation of time when there is too much artificial light at night. Behavior governing mating, migration, sleep, and finding food are determined by the length of night time. Light pollution negatively disrupts these age-old patterns.

The International Dark-Sky Association (IDA), is an educational organization that seeks to preserve the natural night skies worldwide. Light pollution is an increasing problem threatening astronomical facilities, ecologically sensitive habitats, all wildlife, our energy use as well as our human heritage. Light pollution is excessive and inappropriate artificial light. !e four components of light pollution are often combined and overlapping.

Urban sky glow—the brightening of the night sky over inhabited areas.

Light trespass—light falling where it is not intended, wanted, or needed.

Glare—excessive brightness which causes visual discomfort. High levels of glare can decrease visibility.

Clutter—bright

Confusing, and excessive groupings of light sources, commonly found in over-lit urban areas. For example proliferation of clutter contributes to urban sky glow, trespass, and glare.

There are solutions to these issues:

- Shield your outdoor lighting
- Only use the light when you need it
- Use timers and dimmers
- Shut off the lights when you can
- Use only enough light to get the job done
- Use long wavelength light with a red or yellow tint to minimize impact

A shielded light uses less wattage and saves you money. Work with your neighbors and local government to keep the skies natural.

IUCN CATAGORIES OF WILD LIFE

The IUCN Red List Categories and Criteria are intended to be an easily and widely understood system for classifying species at high risk of global extinction. The general aim of the system is to provide an explicit, objective framework for the classification of the broadest range of species according to their extinction risk. However, while the Red List may focus attention on those taxa at the highest risk, it is not the sole means of setting priorities for conservation measures for their protection.

Before 1994 the more subjective threatened species categories used in IUCN Red Data Books and Red Lists had been in place, with some modification, for almost 30 years. Although the need to revise the categories had long been recognized (Fitter and Fitter 1987), the current phase of development only began in 1989 following a request from the IUCN Species Survival Commission (SSC) Steering Committee to develop a more objective approach. The IUCN Council adopted the new Red List system in 1994.

OBJECTIVES OF IUCN

The IUCN Red List Categories and Criteria have several specific aims:

- to provide a system that can be applied consistently by different people;
- to improve objectivity by providing users with clear guidance on how to evaluate different factors which affect the risk of extinction;
- to provide a system which will facilitate comparisons across widely different taxa;
- to give people using threatened species lists a better understanding of how individual species were classified.

THE CATEGORIES

EXTINCT (EX)

A taxon is Extinct when there is no reasonable doubt that the last individual has died. A taxon is presumed extinct when exhaustive surveys in known and/or expected habitat, at appropriate times (diurnal, seasonal, annual), throughout its historic range have failed to record an individual. Surveys should be over a time frame appropriate to the taxon's life cycle and life form.

EXTINCT IN THE WILD (EW)

A taxon is Extinct in the Wild when it is known only to survive in cultivation, in captivity or as a naturalized

population (or populations) well outside the past range. A taxon is presumed Extinct in the Wild when exhaustive surveys in known and/or expected habitat, at appropriate times (diurnal,

seasonal, annual), throughout its historic range have failed to record an individual. Surveys should be over a time frame appropriate to the taxon's life cycle and life form.

CRITICALLY ENDANGERED SPECIES (CR)

A category containing those species that possess an extremely high risk of extinction as a result of rapid population declines of 80 to more than 90 percent over the previous 10 years (or three generations), a current population size of fewer than 50 individuals, or other factors

ENDANGERED (EN)

A designation applied to species that possess a very high risk of extinction as a result of rapid population declines of 50 to more than 70 percent over the previous 10 years (or three generations), a current population size of fewer than 250 individuals, or other factors.

VULNERABLE (VU)

A category containing those species that possess a very high risk of extinction as a result of rapid population declines of 30 to more than 50 percent over the previous 10 years (or three generations), a current population size of fewer than 1,000 individuals, or other factors.

NEAR THREATENED (NT)

A taxon is Near Threatened when it has been evaluated against the criteria but does not qualify for Critically Endangered, Endangered or Vulnerable now, but is close to qualifying for or is likely to qualify for a threatened category in the near future.

LEAST CONCERN (LC)

A taxon is Least Concern when it has been evaluated against the criteria and does not qualify for Critically Endangered, Endangered, Vulnerable or Near Threatened. Widespread and abundant taxa are included in this category.

DATA DEFICIENT (DD)

A taxon is Data Deficient when there is inadequate information to make a direct, or indirect, assessment of its risk of extinction based on its distribution and/or population status. A taxon in this category may be well studied, and its biology well known, but appropriate data on abundance and/or distribution are lacking. Data Deficient is therefore not a category of threat. Listing of taxa in this category indicates that more information is required and acknowledges the

possibility that future research will show that threatened classification is appropriate. It is important to make positive use of whatever data are available. In many cases great care should be exercised in choosing between DD and a threatened status. If the range of a taxon is suspected to be relatively circumscribed, and a considerable period of time has elapsed since the last record of the taxon, threatened status may well be justified.

NOT EVALUATED (NE)

A taxon is Not Evaluated when it is has not yet been evaluated against the criteria.

IUCN categories of threatened species

Wildlife MCQs

1) Which of the following areas do not have 'Himalyan moist temprate forest'?

a) Neelam valley
b) Kaghan valley
c) Gilgit
d) Shogran

2) Height of conifers belong to the temprate zone is:
a) 500-1000m
b) 1650-3000m
c) 1750-3000m
d) 1000-5000m

3) In Himalyan moist temprate forests, precipitation is received as -------- during summer:
a) Snow
b) Monsoon rains
c) Ice caps
d) Glaciers

4) How many zones of 'dry-subtropical habitats' are there?
a) 2
b) 3
c) 4
d) 5

5) In subtropical pine forests, a fairly narrow zone is confined b/w:
a) 500-1650m
b) 900-1650m
c) 800-1650m
d) 700-1650m

6) Which of the following animal is not present in sub-tropical pine forests:
a) Kaleej pheasant
b) Babblers
c) Spotted doves
d) Hyaena

7) In tropical deciduous forests dry & hot conditions are present in:
- **a) Early summer & spring**
- b) Late summer & winter
- c) Early winter & late summer
- d) Both a & b

8) Which of the following animals is present in himalyan moist temprate forests?
- a) Koklas pheasant
- b) Jungle crow
- c) Kaleej pheasants
- **d) Both a & b**

9) Dry subtropical habitat is present b/w:
- a) 500 & 900m
- **b) Sea level & 900m**
- c) Sea level & 800m
- d) Sea level & 1900m

10) Height of tropical deciduous forests is :
- **a) 500-1000m**
- b) 600-1000m
- c) 700-1000m
- d) 400-1000m

11) The jackels are found in Pakistan in:

a) Mountainous areas (b) Forest plantations

(c) Deserted areas **(d) Both (a) and (b)**

12) Maximum height of a jackel is:

(a) 35-40cm (b) 40-45cm

(c) 45-50cm (d) 30-35cm

13) Foxes feeds on:

(a) Rabbits (b) Buds and mices

(c) Fruits **(d) All of these**

14) The length of hind foot of red fox:

(a) 110-150cm (b) 110-150m

(c) 110-150mm (d) None of these

15) The bears have very acute sense of:

(a) Touch **(b) Smell**

(c) Hearing (d) Taste

16) Red or brown bears are found in:

(a) Murree (b) Azad Kashmir

(c) Mardan **(d) Hazara Dist.**

17) The shape of Marten:
a) Rounded **b) Cylindrical** c) elliptical

18) The claws of Otters:
a) Retractile **b) Non retractile** c) Both

19) The anal region of Badgers contain which gland:
a) Scent gland b) Musk gland c) None of these

20) The Martens belong to which mountain ranges:
a) Baluchistan b) Northern areas **c) Both**

21) Body length of Martens
a) 75 b) 85 c) 70

22) Ratel have very much reduced

a) Neck **b) ears pinnae** c) forelimbs

23) Coastal Forest also called…

a) Alpine forest
b) Sand dunes
c) Riverine plain
d) Mangrove forest

24) Revirenie plain included the areas…

a) River indus and river of the Punjab
b) Makran coast
c) Sahara desert
d) Kohistan

25) Within the last 400 years the mammalian species loss from Pakistan are…

a) Two
b) Three
c) Four
d) Five

26) Threatened or near threatened species of mammals showed by IUCN red list….

a) 33
b) 37
c) 38
d) 39

27) Sub-species of mammals threatened or near threatened showed by IUCN red list…

a. 11
b. 12
c. 13
d. 14

28) In the recent decades the species that has gone to extinct in Pakistan…

Indian wild ass
Pea fowl
Quails
Otter

29) Vulnerable species of mammals in IUCN red list…

a. 10 **b) 11** c) 12 d) 13

30) Internationally threaten bird species occuring in pakistan

a. **25** b. 27 c. 29 d. 22

31) The critically endangered bird is

A Siberian crane b. Great Indian bustard

c. Lesser florican d. Black cap

32) internationally threaten species of reptiles occuring in pakistan

a. 7 b. 10 c. 15 d. 17

33) species of plants rare in pakistan

a. 450 **b. 500** c. 557 d. 621

34) Rh is a substance found in red cells of human being:

a.75% b. 80% **c. 85%** d. 90

35) The Himalyan otter differ from Indian otter in having its:

A) External appearance B) Size

C) Colour pattern **D) Nose pad indented like**

37) Which of the animal found throughout the lower Indus riverine system of Punjab and sindh:

A) Hyaenas **B) Indian otter**

C) Himalyan otter D) Wild cat

36) Civets are good climbers:

A) Diurnal and herbivores B) and omnivorous Nocturnal

C) Nocturnal and herbivore D) None of these

37) How many species of civets are found in Pakistan :

A) 5 B) 3 **C) 2** D) 1

38) The voice of hyaenas is like

A) Almost humans B) Cats

C) Dogs D) Civets

39) Civets are related to ……..but differ by their longer head and jaws relatively short legs and bushy tails :

A) Bears B) Dogs

C) Cats D) All of them

40) …… is commonly found throughout riverine jungles and sand dune desert of Punjab and Sindh:

A) Hyaeans **B) Indian civets**

C) Palm civets D) Wild cats